

test3 work, PE and inverse square law, KE, conservation energy ↕

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:22pm

Quiz Instructions

One attempt/ NO MAKEUP. NO EXCEPTION. CLICK WISELY. Use a laptop or a tablet. click wisely
you can not see your answer.



Question 1 5 pts

Suppose a 80 kg cosmonaut is at a distance $2R$ from the center of the Earth. R is the radius of the Earth. His/her weight at that distance is about is



100N



200N



400N



800N



1600N



Question 2 5 pts

A 60-kg skier starts from rest from the top of a 50-m high slope. If the work done by friction is -6.0 kJ (-6,000J), what is the speed of the skier on reaching the bottom of the slope?



24 m/s



17 m/s



28 m/s



31 m/s



Question 3 5 pts

A Miami driver is driving at 20mph and the KE of the car + driver is 1,000,000 J

The driver gets enraged (its Miami..) because someone cuts him off. He speeds up to 60mph, His new KE is:

(very simple Math)



3,000,000J



1,000,000J



9,000,000J



4,000,000J

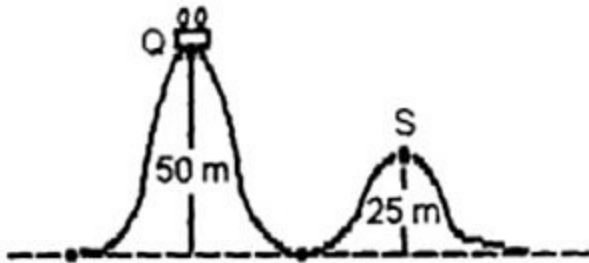


2,000,000J



Question 4 5 pts

The figure shows a famous roller coaster ride. You can ignore friction. If the roller coaster leaves point Q from rest, what is its speed at the top of the 25-m peak (point S)?



62 m/s



44 m/s



10 m/s



120 m/s



22 m/s



Question 5 5 pts

A 30-N stone is dropped from a height of 10 m and strikes the ground with a speed of 13 m/s. What average force of air friction was exerted on the stone as it fell?

hint: take your time. Don't confuse weight and mass. $g=9.8\text{m/s/s}$. consider energy before and after.



1.2 N



4.1 N



7.2 N



2.9 N



0.13 kN



Question 6 5 pts

Starting from rest, a marble first rolls down a steeper hill, then down a less steep hill of the same height. what is true about the speed at the bottom?



Same speed at the bottom of both hills



Faster at the bottom of the less steep hill



Can't say without knowing the mass of the marble



Faster at the bottom of the steeper hill



Question 7 5 pts

Which of the following bodies has the largest kinetic energy?

Mass $3M$ and speed V Mass $3M$ and speed $2V$ Mass $2M$ and speed $3V$ Mass M and speed $4V$

if Algebra is challenging, decide of a mass of $M=1\text{kg}$ for example and a velocity V of 1m/s then do the Math



B



D



C



A



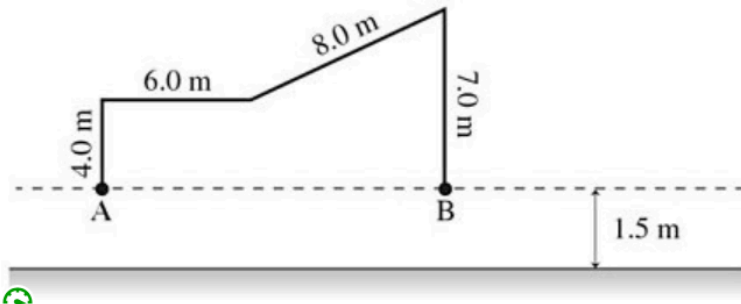
Question 8 5 pts

A bucket of water with a total weight of 50 Newtons is lifted at constant velocity up a 10 meter deep well. If it takes 20 seconds to raise the bucket this distance, the Power required to lift the bucket is

- ☐ 2.5 J
- ☐ 25 W
- ☐ 500 W
- ☐ 50 W
- ☐

Question 9 5 pts

A person carries a 25.0-N rock (the system) through the path shown in the figure, starting at point A and ending at point B. The total time from A to B is 1.50 min. How much work did the gravitational force exerted on the rock by Earth do on the rock between A and B?

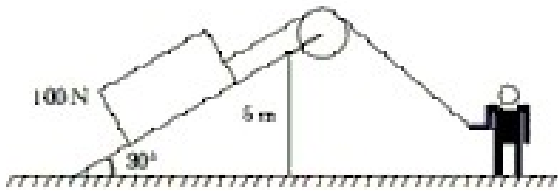


hint: that is the work done by gravity

- ☐ 20.0 J
- ☐ 275 J
- ☐ 0 J
- ☐ 625 J
- ☐ 75 J
- ☐

Question 10 5 pts

A man pulls a 100-N crate up a frictionless 30 degrees slope 5m high, as shown. Assuming that the crate moves at constant speed, the work done by the man is:

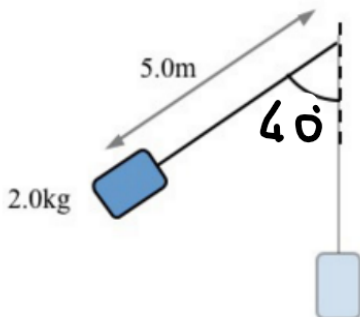


- ☐ -1000 J
- ☐ 1000 J
- ☐ 500 J
- ☐ 250J



Question 11 5 pts

A mass of 2 kg is attached to the end of a light string to make a pendulum 5.0 m in length. The mass is raised to an angle of 40 degrees relative to the vertical, as shown, and released. The speed of the mass at the bottom of its swing is. (round). $g=10\text{m/s/s}$



- ☐ 3 m/s



23 m/s



5 m/s



14 m/s



Question 12 5 pts

100 cm = 1m

A 60.0-kg person drops from rest a distance of 1.20 m to a platform of negligible mass supported by an ideal stiff spring of negligible mass. The platform drops 6.00 cm before the person comes to rest. What is the spring constant of the spring?

take $g=9.8\text{m/s/s}$. !! **its a tricky question. You have to draw first and decide of a reference level for PE . where $PE=0$**

 $2.56 \times 10^5 \text{ N/m}$  $3.92 \times 10^5 \text{ N/m}$  $5.45 \times 10^4 \text{ N/m}$  $4.12 \times 10^5 \text{ N/m}$  $8.83 \times 10^4 \text{ N/m}$ 

Question 13 5 pts

A 3.0-kg block is dragged over a rough horizontal surface by a constant force of 16 N acting at an angle of 37° above the horizontal as shown. The speed of the block increases from 4.0 m/s to 6.0 m/s in a displacement of 5.0 m. What work was done by the friction force during this displacement?

hint: you can use $(\text{change in KE}) = W_{\text{net}}$. forces are weightx and push and friction. weightx is the x-component of the weight along the slope. work is negative obviously

Or you can use conservation of energy.



64J

☐

-30J

☐

-34 J

☐

-64J

⋮

Question 14 5 pts

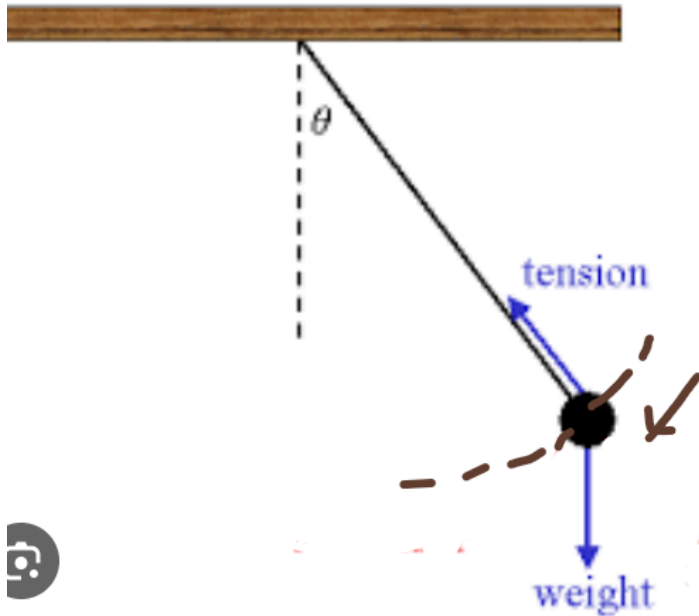
A particle is in a potential $U(x) = 5x^2$. Assume $U(x) = 0$. What is the conservative force experienced by this particle. (use the definition of a conservative force. see slides)

Remember. How the restoring force is defined with the potential energy.

☐ $F = -10x$ ☐ $F = -5/3 x^3$ ☐ $F = 5/3 x^3$ ☐ $F = 10x$

⋮

Question 15 5 pts



A pendulum is swinging back and forth. We neglect friction or air resistance. The mass of the bob is m . The bob is acted upon by the tension T and the weight mg . Was a popquiz.

The restoring force is (along the tangent to the path followed by the bob. forces come in components)

☐ $mg\sin(\theta)$

☐ $mg\tan(\theta)$

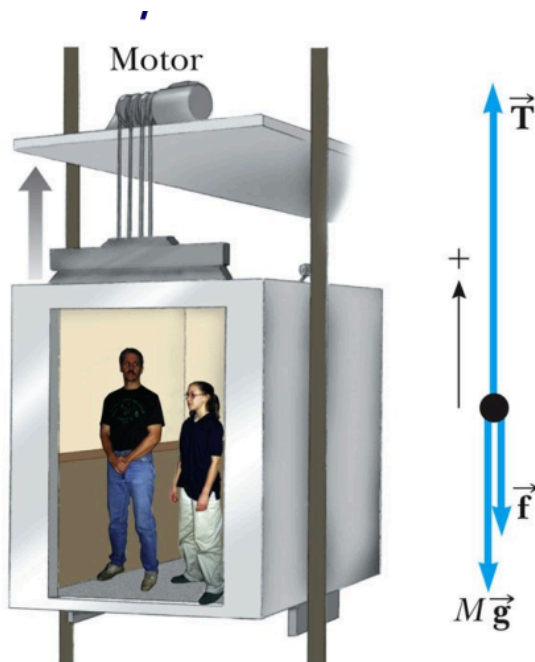
☐ T

☐ $mg\cos(\theta)$

☐ mg

☒

Question 16 5 pts



An elevator has a maximum load of 2000kg. There is a frictional force of 500N. The motor provides the force T (tension). What is the minimum power delivered by the motor if the speed is constant (going up) and is 0.5m/s.

Check your T_____K

☐ about 10kW

☐ about 5kW

☐ about 1kW

☐ about 60kW



Question 17 4 pts



BONUS

The skateboarder swings back and forth on this circular path. Maybe there is friction.

so the forces acting on the kid is: normal force N , weight (mg), kinetic friction f . (draw)

At the bottom (lowest position) what has to be true:



$N=mg$



$N < mg$



$N=0$



$N > mg$



Question 18 5 pts

20. Which force below does the most work? All three displacements are the same.



The 10 N force



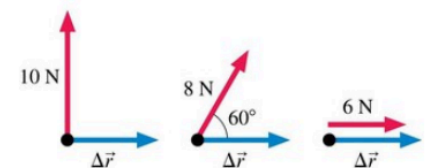
The 8 N force



The 6 N force



They all do the same work.



hint: find the work in each case. slide4 unit6.



C



B



D



A



Question 19 5 pts

30. Two trailers, X with mass 500 kg and Y with mass 2000 kg, are being pulled at the same speed.

The ratio of the kinetic energy of Y to that of X is:



1:1



2:1



4:1



9:1



1500:1

hint: don't rush. you can give yourself a speed like 1m/s. ratio means you divide.



C



D



E



A



B



Question 20 5 pts

Consider the gravitational energy U of a mass m at a distance r from the center of the Earth. (The gravitation energy is 0 at infinity).

Note :Gravitational potential energy is a measure of the energy that a body has due to its position in the gravitational field of the Earth

What is true : U is



is not r dependent and is negative



inversely proportional to the square of r and is negative

☐

is inversely proportional to r and is positive

☐

inversely proportional to r and is negative

☐

proportional to r and is positive



Question 21 5 pts

A roller coaster car is designed to go around a loop-the-loop. The loop has a radius of 10 meters. What is the minimum height from which the car must start to ensure it makes it around the loop without falling off

☐

20m

☐

10m

☐

25m

☐

30m

Not saved

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